

Last-Mile Delivery Exception Handling — Multi-Agent System

Agentic AI · Adewale Adeyinka Falade · LangGraph, LangChain, OpenAI (GPT-4o), SQLite

What It Does

Automatically triages last-mile delivery exceptions (failed attempts, wrong addresses, missed deliveries): reads the delivery log, checks the customer profile, consults a resolution playbook and escalation rules, then produces a resolution decision and a customer-ready message — or escalates to a human when warranted.

Architecture — LangGraph multi-agent graph with self-critique (7 nodes)

1. **Preprocessor** — de-duplicates scans and filters noise (e.g. normal DELIVERED events).
2. **Orchestrator** — routes each case and decides what must happen.
3. **Resolution agent** — proposes a resolution using the tools.
4. **Resolution critic** — reviews and returns weak proposals for revision (reflection loop).
5. **Communication agent** — drafts the customer message.
6. **Communication critic** — checks tone and correctness, looping back if needed.
7. **Finalize** — emits the final structured decision and message.

Tools (5)

read_delivery_logs · lookup_customer_profile · check_locker_availability · search_playbook (retrieval) · check_escalation_rules

Evaluation (a key strength)

Scored against a ground-truth dataset on exception identification, resolution correctness, tone, escalation accuracy and tool-call accuracy. Runs pass these checks with a clean node-by-node trajectory and correct guardrail behaviour (skips tool calls on non-exception events).

Responsible-AI Design

- Guardrails filter noise and prevent unnecessary actions on normal deliveries.
- Escalation rules route the right cases to humans instead of auto-resolving.
- Self-critique loops catch weak resolutions and off-tone messages before they go out.

Self-authored prototype. Customer database, delivery logs, ground-truth and playbook were synthetic course data and are not included. API key removed — set your own OPENAI_API_KEY to run.